CodeCrafters

Build your own Redis

Learn about TCP servers, the Redis protocol and more

Resume Building

This challenge is free until 1 April 2024!

Redis is an in-memory data structure store often used as a database, cache, message broker and streaming engine. In this challenge you'll build your own Redis server that is capable of serving basic commands, reading RDB files and more.

Along the way, you'll learn about TCP servers, the Redis Protocol and more.

Stages

Bind to a port	VERY EASY
Respond to PING	EASY
Respond to multiple PINGs	EASY
Handle concurrent clients	MEDIUM
Implement the ECHO command	MEDIUM
Implement the SET & GET commands	MEDIUM
Expiry	MEDIUM

Replication

Configure listening port	EASY
The INFO command	EASY

The INFO command on a replica	MEDIUM
Initial Replication ID and Offset	EASY
Send handshake (1/3)	EASY
Send handshake (2/3)	EASY
Send handshake (3/3)	MEDIUM
Receive handshake (1/2)	EASY
Receive handshake (2/2)	EASY
Empty RDB Transfer	EASY
Single-replica propagation	MEDIUM
Multi Replica Command Propagation	HARD
Command Processing	HARD
ACKs with no commands	EASY
ACKs with commands	MEDIUM
WAIT with no replicas	MEDIUM
WAIT with no commands	MEDIUM
WAIT with multiple commands	HARD

RDB Persistence

RDB file config	EASY
Read a key	MEDIUM
Read a string value	MEDIUM
Read multiple keys	MEDIUM
Read multiple string values	MEDIUM
Read value with expiry	MEDIUM

Streams

The TYPE command	EASY
Create a stream	MEDIUM

Validating entry IDs	EASY
Partially auto-generated IDs	MEDIUM
Fully auto-generated IDs	MEDIUM
Query entries from stream	MEDIUM
Query with -	EASY
Query with +	EASY
Query single stream using XREAD	MEDIUM
Query multiple streams using XREAD	MEDIUM
Blocking reads	HARD
Blocking reads without timeout	MEDIUM
Blocking reads using $	EASY

> The Redis challenge was extremely fun. I ended up having to read the Redis Protocol specification doc pretty carefully in its entirety! The result felt like lightly-guided independent study, if that makes sense. (Which, again, was lots of fun)

Charles Guo
Software Engineer, Stripe

> I think the instant feedback right there in the git push is really cool. Didn't even know that was possible!

Patrick Burris
Senior Software Developer, CenturyLink

Programming challenges for seasoned developers.

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